CM 8205 สื่อ เทคโนโลยีและวัฒนธรรม

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Related Talks

- **New Digital Media**
- **Interactive Digital Multimedia Applications**
- **Research-based Digital Media for Innovation**
- **Multidisciplinary between Information Technology and Digital Media**

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Recent Computer Vision Applications

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Picture Collage

By Dr. Jian Sun
Visual Computing Group,
Microsoft Research Asia (MSRA)

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What's Picture Collage?

- Inputs:

- How to arrange a set of images on a given canvas?

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Mosaic Collage

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Google's Picasa Collage

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Our Picture Collage

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Picture Collage

- A “visual” image summarization
  - Efficiently view a set of images
  - Maximizing visible visual information
- For image browsing, printing, and sharing
  - Causally images arrangement is easy by hand
  - But it is difficult to the user by mouse

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Learning Realistic Human Actions from Movies

By Dr. Ivan Laptev
INRIA Paris, France
Institut national de recherche en informatique et en automatique

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Learning Realistic Human Actions from Movies

- The aim of this paper is to address recognition of natural human actions in diverse and realistic video settings.
- This challenging but important subject has mostly been ignored in the past due to several problems one of which is the lack of realistic and annotated video datasets.

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Learning Realistic Human Actions from Movies

- Their first contribution is to address this limitation and to investigate the use of movie scripts for automatic annotation of human actions in videos.
- They evaluate alternative methods for action retrieval from scripts and show benefits of a text-based classifier.

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Learning Realistic Human Actions from Movies

- Using the retrieved action samples for visual learning, we next turn to the problem of action classification in video.
- They present a new method for video classification that builds upon and extends several recent ideas including local space-time features, space-time pyramids and multi-channel non-linear SVMs.

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Video result

Learning Realistic Human Actions from Movies

By Dr. Ivan Laptev
INRIA Paris, France

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Flash Matting

By Dr. Jian Sun
Visual Computing Group,
Microsoft Research Asia (MSRA)

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Flash Matting

- In this paper, they propose a novel approach to extract mattes using a pair of flash/no-flash images.
- Their approach, which they call flash matting, was inspired by the simple observation that the most noticeable difference between the flash and no-flash images is the foreground object if the background scene is sufficiently distant.

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- Flash matting results for flower scene with complex foreground and background color distributions.
- From left to right: flash image, no-flash image, recovered alpha matte, and flower basket with new background.

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Method of Flash Matting

- They apply a new matting algorithm called joint Bayesian flash matting to robustly recover the matte from flash/no-flash images.
- Even for scenes in which the foreground and the background are similar or the background is complex

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Video result

Flash Matting

By Dr. Jian Sun from Microsoft

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Drag-and-Drop Pasting

By Dr. Jian Sun

Visual Computing Group,

Microsoft Research Asia (MSRA)

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(a) (b) (c) (d) (e)

- Given a source image (a), the user draws a boundary that circles the wood log and its shadow in the water, and drags and drops this region of interest onto the target image in (b).
- The result from Poisson image editing (c) is however not satisfactory. Structures in this target image (e.g., the dark beach) intersect the source region boundary, thus produce unnatural blurring after solving the Poisson equations.
- Their approach, called *drag-and-drop pasting, computes an optimized boundary shown in (d),* which is then used to generate a seamless image composite (e).

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Video result

Drag-and-Drop Pasting

By Dr. Jian Sun from Microsoft

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Recent Sketching Interface Applications in CG

Dr. Takeo Igarashi,
University of Tokyo, Japan

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Teddy: A Sketching Interface for 3D Freeform Design

- By Dr. Takeo Igarashi, University of Tokyo, Japan
- Teddy is a sketch-based 3D modeling software.
  You can make interesting 3D models just by drawing freeform strokes.

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Teddy (cont.)

- Teddy requires Java installed in your machine, and mainly designed for Windows.
- Commercial Products based on Teddy: PC package , GameCube game , PlayStation2 game

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Dr. Takeo Igarashi, University of Tokyo

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Human Detection

Dr. William R. Schwartz,
University of Maryland, United States

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Human Detection

- Human Detection Using Partial Least Squares Analysis
- Detect the pedestrians robustly

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Dr. William R. Schwartz,
University of Maryland,
United States

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Thank you

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